

Hanwen Zhang

Computer Science Honors Class, Sichuan University | Chengdu, China

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EDUCATION

Sichuan University

Chengdu, China

B.Eng. in Computer Science and Technology

Sep 2024 – Present

- Top-notch Student Training Program 2.0 (Honors Class)
- **GPA: 3.96/4.00 · Rank: 1st in honors class** · Weighted average: 93.76/100 (grades as of September 23, 2026)
- Selected coursework: Operating System (99), Theory of Computation (98), Theory of Optimization (98), Probability Statistics (97), Numerical Computing Method (97), Introduction to Artificial Intelligence (96), Discrete Mathematics (94), Introduction to Multimodal Learning (90)

RESEARCH INTERESTS

Generative 3D vision and video diffusion for incomplete observations; current work focuses on single-image animal reconstruction and sparse-view scene generation.

Longer-term interests include multimodal world models for multiview and future-state prediction, grounded spatial reasoning, controllable scene simulation, and embodied agents.

RESEARCH EXPERIENCE

Bone-to-Shape: Single-Image Animal Reconstruction

Jan 2026 – Present

Project Lead, Provincial Undergraduate Innovation Training Program

Sichuan University

- Lead a project combining AniMer-based animal pose and shape estimation with a ReconX-inspired sparse-view video diffusion stage and 3D Gaussian Splatting (3DGS) reconstruction.
- Generate orbit-view video from a single animal image and study how anatomical and body-shape priors constrain reconstructed geometry.
- Approved January 19, 2026 (applied November–December 2025); passed the college midterm review; first-listed inventor on the related invention patent application.

City University of Hong Kong

Jul 15 – Sep 1, 2026

Research Intern, Department of Data Science (Supervisor: Xinke Li)

Hong Kong

- Remote Jul 15–31; on site Aug 1–Sep 1.
- Explored sparse-view scene generation with video diffusion models.
- Explored Vision Banana as a generative visual model for scene generation.

Deep-Hole Guardian: Embodied Perception for Defect Inspection

Jan 2026 – Present

Team Member, University-Level Undergraduate Innovation Training Program

Sichuan University

- Ran tool-based experiments and analyzed depth point clouds for deep blind-hole defect identification and control with a SCARA robot. Passed the college midterm review.

Kernel-Based Outlier Detection

2026

Research Collaboration

Sichuan University

- Contributed to method design and formal analysis for kernel self-representation learning based fuzzy neighborhood outlier detection; third author of the resulting IJAR article.

PUBLICATIONS

Yijia Chen, Ze Zhang, **Hanwen Zhang**, Jiaqi Yang, Bingye Zhou, Zhong Yuan. Kernel self-representation learning based fuzzy neighborhood outlier detection. *International Journal of Approximate Reasoning*, 198:109769, 2026.
[doi:10.1016/j.ijar.2026.109769](https://doi.org/10.1016/j.ijar.2026.109769)

PATENTS & SOFTWARE COPYRIGHTS

Animal 3D Reconstruction Using Body-Shape Priors

Filed Sep 10, 2026

- **First-listed inventor**; applicant: Sichuan University.
- Chinese invention patent application 202611403381.7. Passed preliminary examination on September 22, 2026; pending publication, not yet granted.

Granular Network Threat Detection System Based on Subspace Learning V1.0

Jul 3, 2025

- Joint copyright holder. Chinese software copyright 2025SR1158898.

HONORS & AWARDS

First-Class Single-Category Scholarship , Sichuan University	2025 – 2026
Third Prize, Sichuan Division , 17th Chinese Mathematics Competition for College Students, Non-Mathematics Category A	2025
Third-Class Comprehensive Scholarship , Sichuan University	2024 – 2025
Outstanding Undergraduate Student Cadre , Sichuan University	2024 – 2025
Provincial Third Prize, Ascend AI Track , Huawei ICT Competition	2024
Successful Participant , Mathematical Contest in Modeling (MCM)	2026
CCF CSP : highest score 250/500 (top 20.6%, Mar 30, 2025); top 11.23% in the May 31, 2026 sitting	2025 – 2026

SOFTWARE PROJECTS

Paper Refiner <i>Python, LaTeX, JSON patches</i> GitHub Multi-agent LaTeX paper revision system. Designed and implemented the workflow and orchestrator, including structured patch-based editing and reproducible runs with configurations and logs.	
Granular Network Threat Detection System <i>Subspace learning, granular computing</i> Co-developed a network threat detection system; joint holder of its registered software copyright.	2025
Smart POS System <i>C++, Qt</i> GitHub Checkout application. Worked on application refactoring, asynchronous database access, UI components, and error handling.	
Bouncing Birds <i>C++, SFML</i> GitHub Physics game. Worked on gameplay, collision handling, saving and loading game state, and replay functionality.	

ACTIVITIES & SERVICE

Sichuan University AI Club <i>President; previously Vice President</i>	
College of Computer Science, Sichuan University <i>League Branch Secretary, 2024 Honors Class</i> Helped organize the class’s successful “Jiang Jie Class” application.	
UNSW Business School <i>Discovery: Foundations and Applications (30-hour in-person program)</i> Led development of a Hackathon prototype combining financial candlestick analysis and multi-agent collaboration.	Jul 20 – 31, 2026
CNCC 2025 <i>Conference participant</i>	Oct 22 – 25, 2025 <i>Harbin, China</i>

SKILLS

Research methods : generative vision; video-diffusion-based novel-view synthesis; sparse-view scene generation; single-image animal pose and shape estimation with AniMer; 3D Gaussian Splatting (3DGS); depth point-cloud analysis	
Machine learning : PyTorch, scikit-learn; kernel self-representation learning; fuzzy-neighborhood outlier detection; method design and formal analysis	
Programming : Python, C/C++	
Research engineering : Linux, Git, Docker, LaTeX; configuration-driven experiments, JSON-based patch workflows, and reproducible logs	
Software frameworks : Qt, SFML	
Languages : Chinese (native); English (CET-6: 538)	